



SAINT IGNATIUS' COLLEGE RIVERVIEW
YEAR 12

MATHEMATICS

Extension One

APRIL
2008

*Time allowed – 2 hours
(plus 5 minutes reading time)*

Directions to Candidates

1. Attempt **ALL** questions.
2. There are **SEVEN QUESTIONS** of equal value.
3. All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
4. Board-approved calculators may be used.
5. Each question attempted is to be returned in a **SEPARATE BOOKLET** clearly marked Question 1, Question 2,etc.
6. Each answer sheet must show your **NAME** and your **TEACHER'S NAME**.

Question One: (12 marks) Please use a separate writing booklet

- a) Solve the following equation. 3

$$|2x + 4| = 9$$

- b) If $\sec \theta = \frac{-5}{4}$ and $\tan \theta > 0$, find the exact value of $\cot \theta + \cos \theta$. 3

- c) Solve the following equation for x : 2

$$4^{2x-1} = \frac{1}{8}$$

- d) An arc of length 2cm subtends an angle of $\frac{\pi}{3}$ at the centre of a circle with radius x cm. Find x . 2

- e) The point A $(-2,1)$ is the midpoint of $(a,4)$ and $(-3,b)$. Find a and b . 2

Question Two: (12 marks) Please use a separate writing booklet

- a) AB is a chord of length 80cm and is 9cm from the centre of the circle. Find the diameter of the circle. 2

- b) Find the following limit. 2

$$\lim_{x \rightarrow 0} \frac{2x}{\sin 5x}$$

- c) Find the size of the angle between the lines $2x - 3y + 2 = 0$ and $x + 2y - 5 = 0$. Answer to the nearest degree. 3

- d) Find the coordinates of the point P(x, y) which divides the interval joining A(6,-4) and B(-1,5) externally in the ratio 2:3. 2

- e) Solve for x and graph the solution on a number line. 3

$$\frac{2x+5}{x+1} \geq 3$$

Question Three: (12 marks) Please use a separate writing booklet

a) Determine the value of 3

i) $\tan^{-1}(\sqrt{3})$

ii) $\cos^{-1}(\sin \frac{\pi}{3})$

b) If $y = \log_e(x - 2)$ 1

i) find the equation of the inverse function

ii) state the domain of this inverse function. 2

c) Draw a neat accurate sketch of the following function. Label clearly any key points. 3

$$y = 4 \sin^{-1} \frac{x}{2}$$

d) The area bounded by the curve $y = \frac{1}{\sqrt{9+x^2}}$, the x axis and the 3
ordinates $x = -3$ and $x = \sqrt{3}$ is rotated about the x axis. Find the
volume of the solid of revolution formed.

Question Four: (12 marks) Please use a separate writing booklet

- a) i) By using the sum to n terms of an Arithmetic Series show that the sum of the first positive n odd integers is n^2 . 1
ie: $1+3+5+\dots+(2n-1)=n^2$
- ii) Prove the result of part i) by Mathematical Induction. 3
- b) Use Mathematical Induction to show that $5^n \geq 1+4n$ for all positive integers n . 4
- c) Prove by Mathematical Induction that $2n + n^3$ is a multiple of 3, for all positive integers n . 4

Question Five: (12 marks) Please use a separate writing booklet

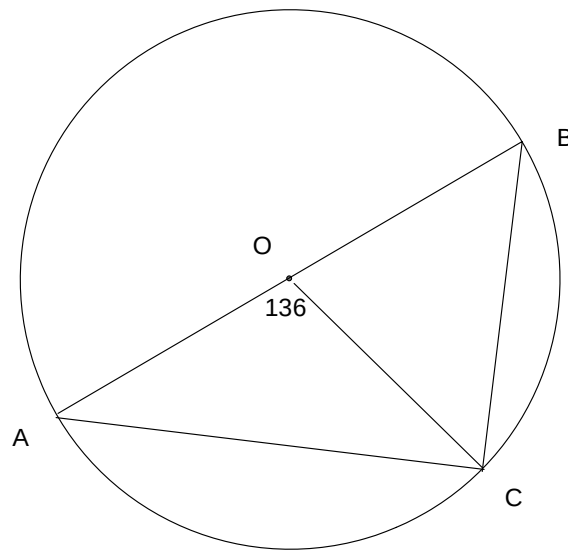
- a) i) Write the expansion of $\sin(\alpha + \beta)$ 1
- ii) Hence, find the exact value of $\sin 105^\circ$ 2
- b) Prove the following identity: $\sin 2B = \frac{2 \tan B}{1 + \tan^2 B}$ 3
- c) Given that $0 < \alpha < \frac{\pi}{4}$, prove that $\tan\left(\frac{\pi}{4} + \alpha\right) = \frac{\cos \alpha + \sin \alpha}{\cos \alpha - \sin \alpha}$ 3
- d) Find all the angles β with $0 \leq \beta \leq 2\pi$ for which $\sin \beta + \cos \beta = 1$ 3

Question Six: (12 marks) Please use a separate writing booklet

- a) Find the cartesian equation of the parabola with parametric equations $x = 6t$ and $y = 2t^2$. 1
- b) Prove that the normal to the parabola $x^2 = 8y$ at the point $P(4t, 2t^2)$ has equation $x + ty = 2t^3 + 4t$ 3
- c) The point $T(t, t^2 + 1)$ lies on the parabola $y = x^2 + 1$. 2
- i) Find the equation of the tangent at T . 1
- ii) The tangent at T and the tangent at the vertex meet at R . Find the coordinates of R . 1
- d) On the parabola $x^2 = y$ the point P has coordinates (t, t^2) . The equation of the tangent at P is $y - 2tx + t^2 = 0$. 1
- i) State the coordinates of the focus S of the parabola. 1
- ii) Prove that the line through S meeting the tangent at right angles in N has equation $4ty + 2x = t$. 2
- iii) Hence determine the equation of the locus of N as P varies on the curve. 2

Question Seven: (12 marks) Please use a separate writing booklet

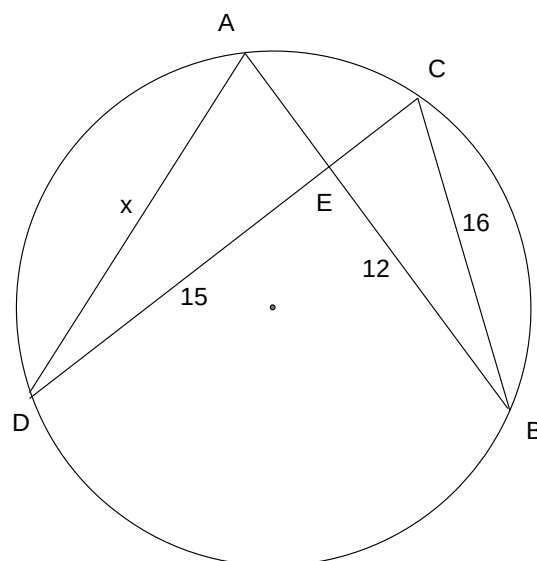
a)



It is known that $\angle AOB$ is straight, $\angle AOC = 136^\circ$ and O is the centre of the circle. Find the size of $\angle OCB$.

3

b)



i) Prove that $\triangle AED$ is similar to $\triangle CEB$

2

ii) Find the value of x

2

c) Two circles touch externally in P . QR is a common tangent to the circles touching the first circle in Q and the second circle in R . The common tangent at P meets QR in S .

i) Draw a clearly labelled diagram of this information.

1

ii) Prove that S is the midpoint of QR

2

iii) Prove that $\angle QPR$ is a right angle

2

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

YR 12 EXT 1 MATHEMATICS.

TERM ONE 2008 EXAMINATION.

SOLUTIONS

QUESTION ONE

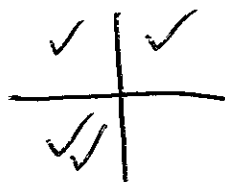
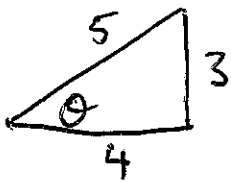
a) $|2x + 4| = 9$

$$2x + 4 = 9 \quad 2x + 4 = -9$$

$$2x = 5 \quad 2x = -13$$

$$x = 5/2 \quad x = -\frac{13}{2}$$

b) $\sec \theta = -\frac{5}{4} \quad \tan \theta > 0$



$$\begin{aligned} \cot \theta + \cos \theta &= \frac{4}{3} - \frac{4}{5} \\ &= \frac{8}{15} \end{aligned}$$

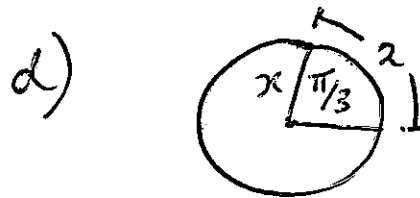
c) $4^{2x-1} = \frac{1}{8}$

$$2^{4x-2} = 2^{-3}$$

$$4x - 2 = -3$$

$$4x = -1$$

$$x = -\frac{1}{4}$$

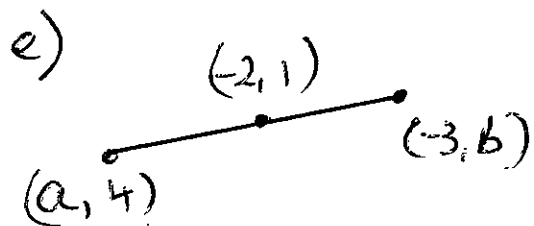


$$l = r \cdot \theta$$

$$2 = x \cdot \frac{\pi}{3}$$

$$\frac{6}{\pi} = x$$

$$x = \frac{6}{\pi} \text{ cm.}$$



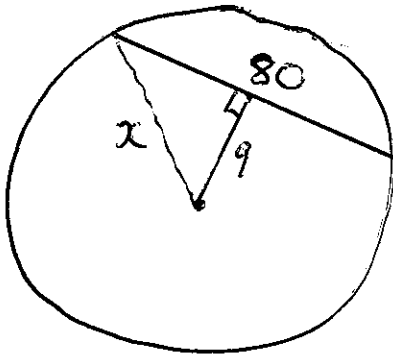
$$-2 = \frac{a + (-3)}{2} \quad 1 = \frac{4 + b}{2}$$

$$-4 = a - 3 \quad 2 = 4 + b$$

$$a = -1 \quad b = -2$$

Question 2

a)



$$x^2 = 9^2 + 40^2$$

$$x = 41$$

$$\therefore \text{DIAMETER} = 82 \text{ cm}$$

b) $\lim_{x \rightarrow 0} \frac{2x}{\sin 5x}$

$$= \frac{2}{5} \lim_{x \rightarrow 0} \frac{5x}{\sin 5x}$$

$$= \frac{2}{5}$$

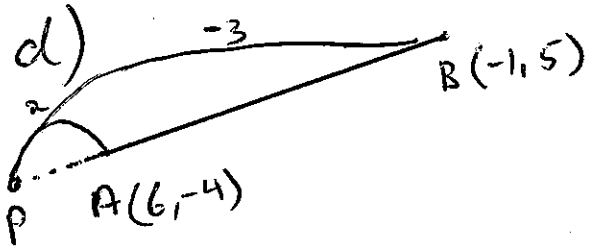
c) $2x - 3y + 2 = 0 \quad m = \frac{2}{3}$

$$x + 2y - 5 = 0 \quad m = -\frac{1}{2}$$

$$\tan \theta = \left| \frac{\frac{2}{3} + \frac{1}{2}}{1 + \frac{2}{3}(-\frac{1}{2})} \right|$$

$$= \frac{7}{4}$$

$$\therefore \theta = 60^\circ 15'$$



$$x = \frac{2(-1) - 3(6)}{-1}$$

$$= 20$$

$$y = \frac{2(5) - 3(-4)}{-1}$$

$$= -22$$

$$P(20, -22)$$

e) $\frac{2x+5}{x+1} \geq 3 \quad x \neq -1$

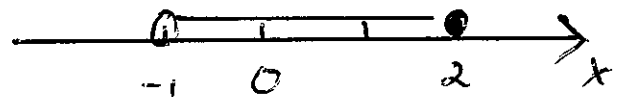
$$(2x+5)(x+1) \geq 3(x+1)^2$$

$$(2x+5)(x+1) - 3(x+1)^2 \geq 0$$

$$(x+1)(2x+5-3(x+1)) \geq 0$$

$$(x+1)(2x+5-3x-3) \geq 0$$

$$(x+1)(2-x) \geq 0$$



Question 3

a) i) $\tan^{-1} \sqrt{3} = \frac{\pi}{3}$

ii) $\cos^{-1} \left(\sin \frac{\pi}{3} \right) = \cos^{-1} \frac{\sqrt{3}}{2}$
 $= \frac{\pi}{6}$

b) $y = \log_e (x-2)$

i) $x = \log_e (y-2)$

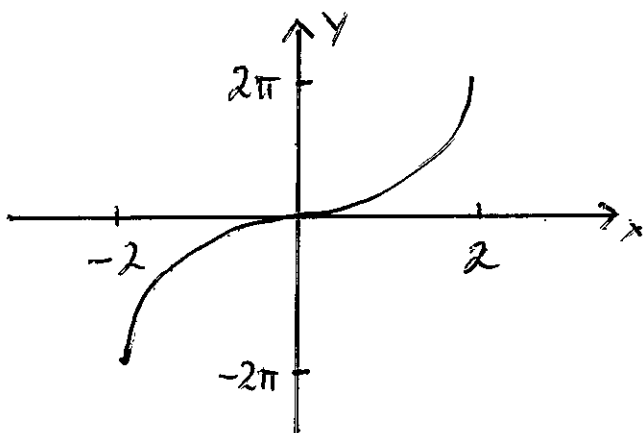
$$e^x = y-2$$

$$y = e^x + 2$$

ii) D: all real x

R: $y \geq 2$

c) $y = 4 \sin^{-1} \frac{x}{2}$



d) $y = \frac{1}{\sqrt{9+x^2}}$

$$V = \pi \int_{-3}^{\sqrt{3}} y^2 dx$$

$$= \pi \int_{-3}^{\sqrt{3}} \frac{1}{9+x^2} dx$$

$$= \pi \left[\frac{1}{3} \tan^{-1} \frac{x}{3} \right]_{-3}^{\sqrt{3}}$$

$$= \frac{\pi}{3} \left[\tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} (-1) \right]$$

$$= \frac{\pi}{3} \left[\frac{\pi}{6} + \frac{\pi}{4} \right]$$

$$= \frac{5\pi^2}{36} u^3.$$

Question 4

a) i) $1 + 3 + 5 + \dots + 2n-1$

$$S_n = \frac{n}{2} (2 + (n-1)2)$$

$$= \frac{n}{2} (2n)$$

$$= n^2.$$

ii)

Prove $1 + 3 + 5 + \dots + 2n-1 = n^2$

Step 1. Prove true for $n=1$

$$\text{LHS} = 1$$

$$\text{RHS} = 1$$

$\therefore$ true for $n=1$

Step 2. Assume true for $n=k$

$$\text{i.e. } 1 + 3 + 5 + \dots + 2k-1 = k^2$$

Prove true for $n=k+1$

$$\text{i.e. } 1 + 3 + 5 + \dots + 2(k+1)-1 = (k+1)^2$$

$$\text{LHS} = 1 + 3 + \dots + 2k-1 + 2(k+1)-1$$

$$= k^2 + 2(k+1)-1$$

$$= k^2 + 2k + 1$$

$$= (k+1)^2$$

$$= \text{RHS}$$

$\therefore$ true for $n=k$ and $n=k+1$

Since true for $n=1$ it must be true for $n=2, 3, \dots$

$\therefore$ true for all positive integers n by M.I.

b) Prove $5^n > 1+4n$

Step 1. Prove true for $n=1$

$$\text{LHS} = 5$$

$$\text{RHS} = 5$$

$\therefore$ true for $n=1$.

Step 2: Assume true for $n=k$

$$\text{i.e. } 5^k > 1+4k.$$

Prove true for $n=k+1$

$$\text{i.e. } 5^{k+1} > 1+4(k+1)$$

$$5^{k+1} = 5(5^k)$$

$$> 5(1+4k)$$

$$> 5 + 20k.$$

$$\therefore 5^{k+1} > 5 + 4k$$

$$\therefore 5^{k+1} > 1 + 4(k+1)$$

$\therefore$ true for $n=k$ and $n=k+1$

Since true for $n=1$ it must be true for $n=2, 3, \dots$

$\therefore$ true for all positive integers n by M.I.

c) Prove $2n + n^3$ is a multiple of 3.

Step 1. Above true for $n = 1$

$$2 + 1 = 3$$

$\therefore$ true for $n = 1$.

Step 2: Assume true for $n = k$

$$\text{i.e. } 2k + k^3 = 3M \text{ (M is an integer)}$$

Prove true for $n = k+1$

$$\begin{aligned} 2(k+1) + (k+1)^3 &= 2k+2+k^3+3k^2+3k+1 \\ &= k^3+2k+3k^2+3k+3 \\ &= 3M+3k^2+3k+3 \\ &= 3[M+k^2+k+1] \end{aligned}$$

$\therefore$ a multiple of 3.

$\therefore$ true for $n = k+1$

Since true for $n = 1$ it must be true for $n = 2, 3, \dots$

$\therefore$ true for all positive integers n by M.I.

Question 5

a) i) $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

ii) $\sin(105) = \sin 45 \cos 60 + \cos 45 \sin 60$
 $= \frac{1}{\sqrt{2}} \cdot \frac{1}{2} + \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2}$
 $= \frac{1 + \sqrt{3}}{2\sqrt{2}}$
 $= \frac{\sqrt{2} + \sqrt{6}}{4}$

b) $\sin 2\beta = 2 \sin \beta \cos \beta$
 $= 2 \frac{\sin \beta}{\cos \beta} \cdot \cos^2 \beta$
 $= 2 \tan \beta \cdot \frac{1}{\sec^2 \beta}$
 $= \frac{2 \tan \beta}{\sec^2 \beta}$
 $= \frac{2 \tan \beta}{1 + \tan^2 \beta}$

c) $\tan\left(\frac{\pi}{4} + \alpha\right) = \frac{\tan \frac{\pi}{4} + \tan \alpha}{1 - \tan \frac{\pi}{4} \tan \alpha}$
 $= \frac{1 + \frac{\sin \alpha}{\cos \alpha}}{1 - \frac{\sin \alpha}{\cos \alpha}}$
 $= \frac{\cos \alpha + \sin \alpha}{\cos \alpha} \cdot \frac{\cos \alpha}{\cos \alpha - \sin \alpha}$

$$= \frac{\cos \alpha + \sin \alpha}{\cos \alpha - \sin \alpha}$$

d) $\sin \beta + \cos \beta = 1$

$\tan \frac{\theta}{2} = t$

$$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1$$

$$2t + 1 - t^2 = 1 + t^2$$

$$2t^2 - 2t = 0$$

$$t(t-1) = 0$$

$$t = 0$$

$$t = 1$$

$$\tan \frac{\theta}{2} = 0$$

$$\tan \frac{\theta}{2} = 1$$

$$\frac{\theta}{2} = 0, \pi$$

$$\frac{\theta}{2} = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\theta = 0, 2\pi$$

$$\theta = \frac{\pi}{2}, \frac{5\pi}{2}$$

$\therefore$ Solutions

$$\theta = 0, \frac{\pi}{2}, 2\pi$$

Question 6

a) $x = 6t$ $y = 2t^2$

$$\frac{x}{6} = t$$

$$y = 2 \left(\frac{x}{6} \right)^2$$

$$y = \frac{x^2}{18}$$

b) $x^2 = 8y$ $(4t, 2t^2)$

$$y = \frac{1}{8}x^2 \quad M_T = \frac{1}{4}(4t)$$

$$y' = \frac{1}{4}x \quad M_N = \frac{t}{-\frac{1}{4}}$$

$$y - 2t^2 = -\frac{1}{t}(x - 4t)$$

$$ty - 2t^3 = -x + 4t$$

$$x + ty = 2t^3 + 4t$$

c) i) $y = x^2 + 1$ $T(t, t^2 + 1)$

$$y' = 2x$$

$$M_T = 2t$$

$$y - (t^2 + 1) = 2t(x - t)$$

$$y - t^2 - 1 = 2tx - 2t^2$$

$$y = 2tx - t^2 + 1$$

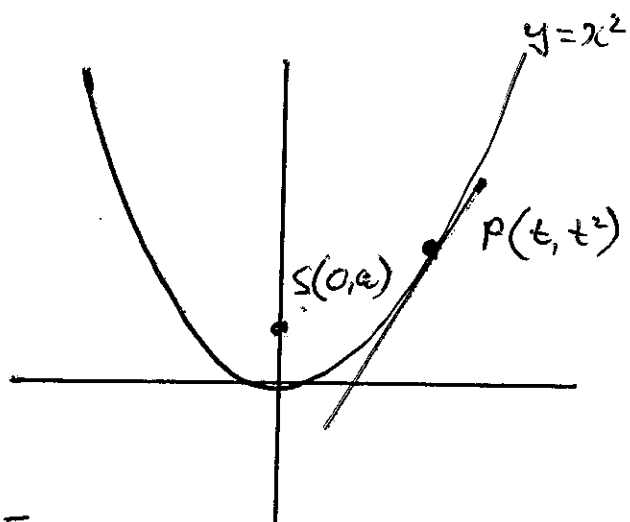
ii) $y = 1$ TANGENT AT VERTEX

$$1 = 2tx - t^2 + 1$$

$$2tx = t^2$$

$$x = \frac{t}{2} \quad R\left(\frac{t}{2}, 1\right)$$

d)



i) Focus

$$x^2 = y$$

$$x^2 = 4\left(\frac{1}{4}\right)y$$

$$\therefore a = \frac{1}{4}$$

$$\text{Focus } \left(0, \frac{1}{4}\right)$$

ii) $y = x^2$

$$y' = 2x$$

$$M_T \text{ at } P = 2t$$

$$M \perp \text{ through } S = -\frac{1}{2t}$$

$$y - \frac{1}{4} = -\frac{1}{2t}(x)$$

$$2ty - \frac{t}{2} = -x$$

$$4ty + 2x = t$$

iii) $y - 2tx + t^2 = 0$ — ①

$$4ty + 2x = t$$
 — ②

From ①

$$y = 2tx - t^2$$
 — ③

Sub into ②

$$4t(2tx - t^2) + 2x = t$$

$$8t^2x - 4t^3 + 2x = t$$

$$x = \frac{4t^3 + t}{8t^2 + 2}$$

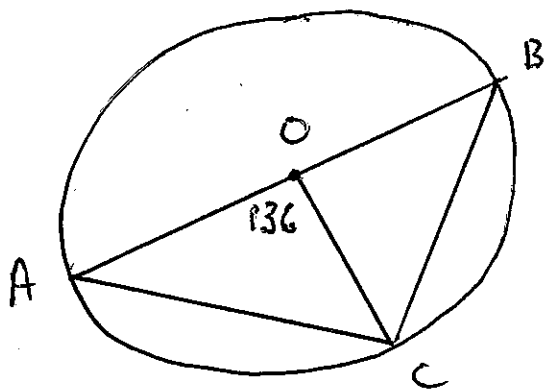
$$= \frac{t}{2}$$

when $x = \frac{t}{2}$ $y = 2t\left(\frac{t}{2}\right) - t^2$

$$\text{Locus } y = 0$$

Question 7

a)



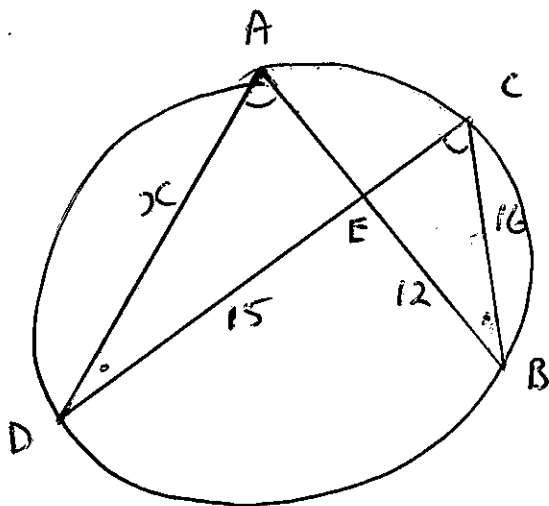
$$\angle OCB + \angle OBC = 136^\circ$$

(Ext $\angle$ of Δ)

$$\angle OCB = \angle OBC \text{ (}\Delta OBC \text{ is isos)}$$

$$\therefore \angle OCB = 68^\circ$$

b)



$$\angle DAB = \angle DCB \text{ (angles on same arc)}$$

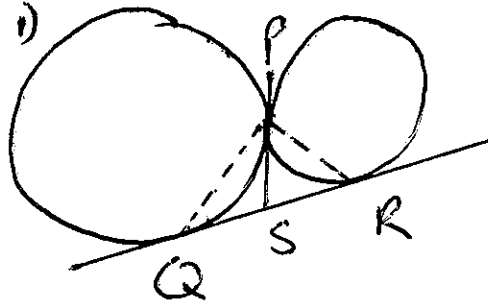
$$\angle ADC = \angle ABC \text{ (angles on same arc)}$$

$$\therefore \Delta AED \parallel \Delta CEB \text{ (equiangular)}$$

$$\therefore \frac{x}{16} = \frac{15}{12} \text{ (Corresponding sides in similar triangles)}$$

$$x = 20$$

c) i)



$$ii) PS = SQ \text{ (Tangents from an external pt)}$$

$$PS = SR \text{ (")}$$

$$\therefore SQ = SR$$

$$\therefore S \text{ is midpoint of } QR$$

$$iii) \text{ Let } \angle SRP = x$$

$$\therefore \angle SPR = x \text{ (}\Delta SPR \text{ is isos)}$$

$$\text{Let } \angle SQP = y$$

$$\therefore \angle SPQ = y \text{ (}\Delta SQP \text{ is isos)}$$

$$x + x + y + y = 180^\circ \text{ (angle sum of } \Delta \text{)}$$

$$\therefore x + y = 90$$

$$\therefore \angle QPR = 90^\circ$$